STANES SCHOOL

1044, Avinashi Road, Coimbatore – 18.

PREPARATORY EXAM – 1 (2026-2027)

Class : XII **Duration : 2 Hours**

Subject : Informatics Practices (065) **Max. Marks : 50**

Section-A (14 × 1 = 14 Marks)

1. Which SQL function is used to extract a specified number of characters from the right side of a string?

a) LEFT() b) RIGHT() c) MID() d) INSTR()

2. What does URL stand for?

a) Uniform Resource Locator b) Universal Reference Link

c) Uniform Reference Locator d) Universal Resource Link

3. Which SQL function is used to display the month name (e.g., January, February) from a date?

a) MONTH() b) MONTHNAME() c) DATE() d) DAYNAME()

4. What is a cookie in the context of web browsing?

a) A virus that infects the computer b) A small text file stored on the user's computer by a website

c) A type of web browser d) A programming language used for web development

5. Which SQL function is used to remove leading spaces from a string?

a) RTRIM() b) TRIM() c) LTRIM() d) LENGTH()

6. What will be the output of the following SQL query? **SELECT MOD(29, 4);**

a) 7 b) 1 c) 7.25 d) 0

7. State whether the following statement is True or False:

A webpage is a collection of related webpages, while a website is a single document on the World Wide Web.

8. Which type of network is formed when a smartwatch connects to a smartphone via Bluetooth?

a) LAN b) MAN c) PAN d) WAN

9. Which SQL clause is used to sort the result set in descending order?

a) GROUP BY b) ORDER BY ... DESC

c) SORT BY d) HAVING

10. Which of the following is a real-time text communication service over the Internet?

a) Email b) Chat c) FTP d) WWW

11. Which of the following SQL functions will convert the string 'DATA' to 'data'?

a) UCASE('DATA') b) UPPER('DATA')

c) LCASE('DATA') d) SUBSTRING('DATA')

12. Which of the following is an example of a web browser?

a) Google b) Mozilla Firefox c) Windows 10 d) Microsoft Word

Q13 and Q14 are Assertion (A) and Reason (R) based questions. Mark the correct choice as:

a) Both A and R are True and R is the correct explanation for A.

b) Both A and R are True and R is not the correct explanation for A.

c) A is True but R is False.

d) A is False but R is True.

13. Assertion (A): A router is used to connect two different networks and forwards data packets between them based on their IP addresses.

Reason (R): A router operates only within a single network and cannot connect two different networks together.

14. Assertion (A): The WHERE clause is used to filter rows before grouping, while the HAVING clause is used to filter groups after grouping.

Reason (R): Both WHERE and HAVING can be used interchangeably in a SQL query with GROUP BY.

Section-B (4 × 2 = 8 Marks)

15. A. Differentiate between DELETE and DROP commands in SQL with suitable examples.

OR

B. Explain the concepts of MAN and WAN with a suitable example for each.

16. Write the output of the following SQL queries based on the table **GROCERY**:

Item_ID	Item_Name	Price	Qty
G201	Rice	60	50
G202	Sugar	45	30
G203	Oil	120	20
G204	Tea	250	10
G205	Salt	25	100

I. SELECT SUM(Price), AVG(Price) FROM GROCERY;

II. SELECT COUNT(*) FROM GROCERY WHERE Price > 50;

17. Differentiate between a static web page and a dynamic web page with suitable examples.

18. Green Valley School is setting up its new computer lab with 40 computers, a printer, and a scanner. The school also wants to connect this lab to the administrative office located 80 meters away, and eventually to a branch campus in another city.

Based on the above scenario, answer the following:

I. Which type of network will be formed within the computer lab?

II. Which network device will you suggest to connect all computers within the lab, and why?

Section-C (4 × 3 = 12 Marks)

19. A. Explain the terms web server and web hosting. Also, state one difference between them.

OR

B. What are browser add-ons and plug-ins? Explain with one example of each.

20. Riya visits an online shopping website. She notices that the website remembers her login details, language preference, and items in her cart even after she closes the browser.

Based on the above scenario, answer the following:

I. What are these small files called that store such information?

II. Name any two commonly used web browsers.

III. What is the purpose of a web server in this context?

21. Consider the table **PATIENT**:

PNo	PName	Dept	DOA
201	Kavya Nair	Cardiology	2024-06-10
202	Rohan Gupta	Neurology	2025-01-15
203	Ishaan Mehta	Cardiology	2023-11-20
204	Diya Sharma	Orthopedics	2022-08-05
205	Arjun Reddy	Cardiology	2025-02-28

Write SQL queries to perform the following tasks:

I. Display the names of all patients in the Cardiology department in uppercase.

II. Display each patient's number along with the month name of the date on which they were admitted.

III. Display the patient names arranged in reverse alphabetical order.

22. Consider a table named **EMPLOYEE** containing the following data:

ID	Name	DOB
1	Karan	2004-07-20
2	Meera	2005-03-12
3	Nikhil	2006-11-05

Write the output of the following SQL queries:

I. SELECT LOWER(Name), LENGTH(Name) FROM EMPLOYEE WHERE ID = 2;

II. SELECT MONTH(DOB), DAY(DOB) FROM EMPLOYEE WHERE ID = 3;

III. SELECT LEFT(Name, 2) FROM EMPLOYEE WHERE ID = 1;

Section-D (3 × 4 = 12 Marks)

23. Consider the table **BOOKS** and **AUTHORS** given below:

Table: BOOKS

Book_ID	Title	Author_ID	Genre	Price
B301	The Alchemist	A01	Fiction	350
B302	Sapiens	A02	Non-Fiction	600
B303	Harry Potter	A03	Fiction	800
B304	Wings of Fire	A04	Autobiography	400
B305	1984	A05	Fiction	500
B306	The Hobbit	A03	Fiction	700

Table: AUTHORS

Author_ID	Author_Name	Country
A01	Paulo Coelho	Brazil
A02	Yuval Noah Harari	Israel
A03	J.K. Rowling	UK
A04	A.P.J. Abdul Kalam	India
A05	George Orwell	UK

Write SQL queries for the following:

I. To display the Title, Author_Name, and Price of all Fiction books using an equi-join.

II. To display the Country and the total number of books written by authors from each country.

III. To display the Title and Author_Name of books that have a price greater than the average price of all books.

IV. To display the Title, Author_Name, and Price of books sorted by Author_Name in ascending order and Price in descending order.

24. CareWell Hospital is setting up its new campus in Chennai with four blocks: Emergency, Surgery, Diagnostics, and Pharmacy. The distances between blocks and the number of computers in each block are given below:

Distances between blocks:

From	To	Distance (in meters)
Emergency	Surgery	60
Emergency	Diagnostics	100
Emergency	Pharmacy	140
Surgery	Diagnostics	80
Surgery	Pharmacy	90
Diagnostics	Pharmacy	50

Number of Computers in each block:

Block	Number of Computers
Emergency	80
Surgery	45
Diagnostics	120
Pharmacy	25

I. Suggest the most suitable block to install the server. Justify your reason.

II. Draw a suitable cable layout of connections between the blocks in the campus.

III. Which hardware device will you suggest to connect all the computers within each block?

IV. Which network topology is most suitable for connecting the blocks in this campus? Justify your answer.

25. Consider the table **STUDENT** and **SCORES** given below:

Table: STUDENT

Roll_No	Name	Class	City
101	Riya	12	Delhi
102	Arjun	11	Mumbai
103	Sneha	12	Chennai
104	Vikram	11	Delhi
105	Pooja	12	Mumbai

Table: SCORES

Roll_No	Subject	Score
101	Mathematics	88
101	Physics	76
102	Chemistry	90
103	Mathematics	95
104	Biology	82
105	Computer Science	91
102	Mathematics	79

Predict the output of the following SQL queries:

I. SELECT Name FROM STUDENT WHERE Class = '12' ORDER BY Name ASC;

II. SELECT DISTINCT UPPER(Subject) FROM SCORES WHERE Score > 80;

III. SELECT STUDENT.Name, SCORES.Subject, SCORES.Score FROM STUDENT, SCORES WHERE STUDENT.Roll_No = SCORES.Roll_No;

IV. SELECT STUDENT.Name, SUM(SCORES.Score) AS Total_Score FROM STUDENT, SCORES WHERE STUDENT.Roll_No = SCORES.Roll_No GROUP BY STUDENT.Name HAVING COUNT(SCORES.Subject) > 1;

V. SELECT Subject, AVG(Score) AS Average_Score FROM SCORES GROUP BY Subject;